

Casper — Idea document

Working name: **Casper**

Status: idea / brainstorm (not a repo yet)

Purpose of this file: capture product, business and technology so it can later be used as input to Cursor when a real repo is created.

This is a living document. Open questions are marked with **TBD**.

1. One-liner

Casper is a **chat-first app** with two tabs: **Connect** (groups, private chats, and event bubbles that organise social play) and **Play** (a session chat that runs the night on court). Pickleball first. Formats are generic enough for other racket sports later.

AI is on demand (Ask Casper / Create Event / run Play, including **register score to the match on screen**). Backend code owns formats and standings. **The person who uses AI pays** — 3 months free, then **49 kr/month**, cancel anytime. Players in a group or session can **read** without paying.

Stack: **AWS** (Cognito, Lambda, DynamoDB, ...) so it can grow to US and Asia later.

2. Problem

Social racket sports already happen in WhatsApp / Messenger groups. The painful part is not “finding a chat”. It is:

1. **Connect** — who is in tonight? Who dropped? Who is standing in the parking lot right now?
2. **Play** — we have N players, C courts, T minutes. Make a fair format *now*. Then keep score and show a table without a spreadsheet on someone’s phone.

Existing tournament tools assume a planned event: fixed draw, named teams, DUPR IDs collected days ahead. They break when:

- 14 people said yes, 11 showed up, 2 more walk in
- one court is taken by a lesson
- someone has to leave after 40 minutes
- the organiser is also playing and has no laptop

Casper is built for that moment: **minutes before, and during, social play** — and for the groups that lead up to it (including focused ones like “Skill 4+ Sundays 9–12”).

We do **not** bolt Casper onto WhatsApp. We build our own chat, and we **share out** (invite links, event cards, recaps) to other messengers so groups can still recruit there.

3. Who it is for

Primary (v1)

- **Organiser (paid / trial)** — has a Casper membership and a role that may use AI. Creates events, asks Casper, starts Play, registers scores in the Play chat.
- **Group creator / admin** — creates groups, invites, toggles quiet mode, assigns who may organise. Can always post in the group. Often the same person as the organiser.
- **Player (free)** — member of a group or participant in a Play session. Can read Connect/Play they belong to, **RSVP**, **like** an event, use **private chats**. Does not pay. Does not invoke AI.

Secondary (later)

- Other racket sports using the same formats (padel, badminton, table tennis)
- DUPR-rated social nights
- Halls / clubs as organisations (not required for v1)

Sport: v1 is **pickleball**. Every event/session has **sport: "pickleball"**. Formats themselves are sport-agnostic. Adding padel later should be a sport tag + scoring defaults, not a rewrite.

Assumption to validate: first users are Swedish / Nordic pickleball groups. **TBD:** Swedish UI vs English-first product with Swedish as a locale.

4. Product shape: two tabs, both chat

```
Casper
├── Tab: Connect
│   ├── Groups          e.g. "Skill 4+ Sundays 9–12" (quiet mode on/off)
│   └── Private chats    1:1 – always available, not affected by group
├── quiet mode
│   └── In a group: messages + **event bubbles** (RSVP, like)
└── Tab: Play
    └── Session chats    from an event, or scratch (no event)
```

Inbox is split by **job**: organising vs running the night.

4.1 Groups and private chats

Someone **creates a group** (a digital club, or a slice of one). Then people **join** (link / code) or creator/admins **invite**.

Groups can be narrow on purpose:

- "Skill 4+ Sundays 9–12"
- "Thursday social — Hall X"
- "Beginners Wednesday"

Invite the right people to the right group. **Weekly events** live in that group as repeating event bubbles.

Private chats sit in the same Connect tab: 1:1 between members. Quiet mode on a *group* does **not** turn off DMs. People should still be able to message each other privately and RSVP on events.

Casper is not sitting in every DM. **TBD:** Ask Casper from a private chat vs Play tab only.

This is not “a hall’s official club” unless they choose to be. It is **groups of people who play together**.

4.2 Roles, membership, and who may use AI

Two gates, both required to create events / Ask Casper / start Play / register scores via AI:

- 1. **Group role** that is allowed to organise (creator, admin, or an “organiser” role)
- 2. **Personal Casper membership** (trial or paid) — **AI use is what you pay for**

	No membership (free player)	Trial / paid membership
Member	Read group + events they are in. RSVP. Like event. Private chats. View Play if participant.	Same; may organise only if the group role allows it
Creator / admin / organiser role	Can read and post (admins); cannot call Casper until trial/pay	Create event, Ask Casper, start Play, register scores via AI

Players do not pay. They can read Connect and Play threads they belong to.

If the organiser stops paying, AI actions stop. Existing groups, events, and Play records stay **readable**. **TBD:** exact grace behaviour.

4.3 Connect — Casper on tap; quiet mode is a group switch

Casper is **not** always listening. Organiser starts it with **Ask Casper** or **Create Event**.

Events **live in the chat as bubbles**: date, time, place, courts, RSVP counts, likes, share, Start Play.

Players on an event bubble can always:

- **RSVP** (in / out / maybe)
- **Like** the event (optional, lightweight interest)

Weekly events: a group like “Skill 4+ Sundays 9–12” can post the next bubble on a schedule.

Quiet mode (creator toggles on/off)

Per **group**, not a global app setting. Creator turns it on so the group is not used as a second WhatsApp.

	Quiet off	Quiet on
Creator / admin post in the group	Yes	Yes, always
Member free-chat in the group	Yes	No (or strongly limited)
Member RSVP	Yes	Yes
Member like event	Yes	Yes
Private 1:1 chats	Yes	Yes — unchanged

So: DMs and RSVP (and likes) stay. The noisy *group* thread is what quiet mode shuts down. Admins can still announce ("court 3 is wet, we start 15 min late").

4.4 Play — from event or scratch

Play is a **new chat**, created when someone with role + membership starts a session.

Two entry points:

- 1. **From an event bubble** — roster copied from Connect. Last-minute drop/add still works.
- 2. **From scratch — allowed.** No Connect event. Organiser starts a format with the people who are here. Always an **organiser**. **Participants can view.**

Play membership = **organiser + participants of this session**, not the whole Connect group.

Job of Play:

- AI interviews the organiser: courts, names, DUPR or not, format, scoring, time
- Backend **generates** the session (catalogue only)
- Live table and next games in the Play chat
- Swap who is actually here
- **Register scores via AI** (see below)
- End session → stored on the organiser; participants can view

Scores (open, with a chosen AI path)

Decided for the AI path: the organiser asks Casper to register a score **to the right match**, because the Play chat **shows on screen which match it is**. The model must bind "11–8" to that match (court / round / pairing in context), then call the backend **record_result**. It must not invent a match.

Still open: may a **player** also tap or type a score, or only the organiser via AI?

Standings and next round after a recorded result are **engine**, not LLM.

During play also: substitute, late arrival, someone leaving, pause/drop a court, undo last score.

4.5 Data ownership and visibility

Thing	Stored / owned	Who can see
Group	Creator / admins	Members
Private chat	The two users	Those two
Connect messages + event bubbles	The group	Members (read). RSVP/like: members. AI: organiser with membership
Event	Organising person; usually a group	Group members
Play session + Play chat	Organising person	Organiser + participants can view

Thing	Stored / owned	Who can see
Match history	Under organiser, linked to players	Organiser; each participant for sessions they played

Scratch Play: `sourceEventId` is null. Still `organiserId` + `participantIds`.

4.6 Sharing (out), not bridging (in)

Own chat. No WhatsApp Business bridge.

Share: group invite, event bubble link, Open in Casper, optional recap image.

4.7 What stays where

Connect: groups, DMs, event bubbles, weekly rhythm, quiet groups.

Play: courts, table, AI score registration, swaps, recap — including scratch sessions.

5. Experience principles

1. **Two chats, two jobs.** Connect organises people. Play runs the night.
2. **Events are bubbles in a group**, not a separate admin website.
3. **Casper speaks when asked** (Ask Casper / Create Event / Play setup / register score).
4. **Quiet mode is a group toggle.** Admins still post. Members still RSVP, like, and DM.
5. **AI use is the paid act.** Readers (players) stay free.
6. **Scratch Play is first-class.** No event required. Organiser + participants can view.
7. **Score goes to the match on screen.** AI + tool call, then engine updates the table.
8. **AI never invents a format** (or a match that is not in the current round).
9. **Last-minute is first-class.**
10. **Deterministic format engine.** Testable.
11. **History is queryable** from stored sessions, scoped by organiser + participation.
12. **Share out, stay in.**
13. **Ordinary Connect messages do not hit OpenAI.** Play score registration does, by design.

6. Formats (backend-owned)

Formats are **versioned code** behind an API. The model may only choose from this catalogue and pass parameters.

Formats are **not pickleball-specific**. Scoring defaults may be (`sport: pickleball` → games to 11, win by 2, doubles).

Proposed v1 catalogue:

Code	Name	When it fits
<code>americano</code>	Americano	Mixed partners, rotating, very common social

Code	Name	When it fits
<code>mexicano</code>	Mexicano	Similar rotation, winners move up
<code>round_robin</code>	Round robin (fixed pairs or rotating)	Smaller groups; also scratch Play
<code>king_of_court</code>	King of the court	Winners stay, losers rotate
<code>social_rotate</code>	Timed rotate	"Play 12 min, rotate" — no ranking required

Later: ladder / box league, single elimination, DUPR constraints on top of a format.

Format engine contract (sketch)

Backend, not the LLM:

- Input: `sport`, `format`, `playerIds[]`, `courts`, `options`
- Output: `rounds[]` → `matches[]` (court, side A, side B, status)
- Commands: `record_result`, `swap_player`, `add_player`, `remove_player`, `rebalance`, `next_round`
- Invariants: no duplicate simultaneous play, court count respected, bye rules explicit

`record_result` is what Casper's tool must call after the organiser names a score. The engine validates match id, score shape, and whose turn it is.

If the roster changes, **rebalance** is a real algorithm.

7. DUPR

Optional per event/session. Relevant while `sport: pickleball`.

When **rated = true**: collect DUPR names/IDs if we have them (**TBD**: official API vs manual), store exportable results. Casper is not DUPR.

When **rated = false**: skip IDs, still store Casper history.

TBD: launch feature vs v2 for clubs?

8. Example evening (happy path)

```
CONNECT – group "Skill 4+ Sundays 9–12" (quiet ON)
          24 members, weekly event
          members cannot free-chat; they RSVP / like; admins can post
          private DMs still work

Creator   weekly event bubble already there
Event bubble: Sun 17 Aug 09:00–12:00, 4 courts, not DUPR
          [I'm in] [Out] [Like] [Share]
...       16 tap in over the week
```

Admin post: "Hall opens 08:45, bring indoor shoes."
Organiser shares the bubble link to an old WhatsApp thread

PLAY – new chat "Sun 17 Aug – Skill 4+"
organiser + participants

09:02 Organiser: We're 11 here, 3 courts. Americano.
Casper: Games to 11? Win by 1 or 2? Until 12:00?
Organiser: 11, win by 1, until 12.
Casper: Starting Americano, pickleball, 11 players, 3 courts, 1 bye.
Round 1:
Court 1 Cassie/Erik vs Anna/Moa ← on screen
Court 2 ...
Bye: Kim

09:14 Organiser (Ask Casper): Court 1 11–8
Casper: Records 11–8 on Court 1 Cassie/Erik vs Anna/Moa (match in view).
Table updated. Next games posted. ← engine after the tool call

09:40 Organiser: Lisa arrived, Kim has to go
Casper: Swapped. Remaining rounds rebuilt.

Scratch (no event): Play → New session → add people → same interview. Organiser owns it; participants can view.

9. Business model

Chosen: organiser pays for AI. Players free. 3 months free, then **49 kr/month**, cancel anytime.

How it works

Who pays	The person who uses AI (Ask Casper, Create Event, run Play, register scores via AI). Not the players.
Who is free	Players: read, RSVP, like, DMs, view Play they participated in.
Trial	3 months free , then paid.
Paid	49 kr / month. Stop anytime — no binding period.
Price job	Cover OpenAI + AWS + ops , small margin. Shrug-and-pay for a Thursday captain.
Gate	AI requires role + active trial/paid membership .

Why not charge players

They already paid for the court. A paywall on "I'm in" kills the group.

Cost control (49 kr must work)

- Connect group chat does **not** go to OpenAI
- Format generation and standings = **Lambda / engine**
- Play interview, swaps, **score registration** = OpenAI tool calling (more calls than “scores are buttons only”)
- Prefer a **cheap model** for **record_result** (short prompt: current matches on screen + “Court 1 11-8”)
- Soft cap **TBD** if someone hammers Ask Casper
- AWS serverless (pay per use) so infra stays small until US/Asia

Rough night: 1 setup + ~15–40 score calls + a few swaps. With a mini model, token cost should sit well under 49 kr. Revisit if we use a large model for every 11–8.

Price details

- **49 kr/month** (working number, Sweden first)
- **TBD:** VAT, Stripe on web vs App Store / Google Play cut, EUR for later markets
- Membership is **per person who uses AI**, not per group. Two captains = two × 49 kr if both Ask Casper. **TBD** group seat later.

When they cancel

- No new AI
- They and players keep **reading**
- **TBD:** resume anytime; trial only once per account

Rejected / later

Model	Status
Per-session fee at the door	Rejected
Every player pays	Rejected
Hall booking bundle	Later
Sponsorship	Ignore for now

Positioning

“The chat that organises the group — and runs club night.”

Players free. Captains: three months free, then 49 kr/month.

10. Technology — AWS first

Familiar ground, and the same platform scales to **US and Asia** later (CloudFront, regional APIs, DynamoDB global tables). We do **not** start on MongoDB Atlas or a random VM.

Primary region (v1): `eu-north-1` (Stockholm) — GDPR, Swedish users. Edge: CloudFront (global).
Later: extra regions + global tables, not a rewrite.

10.1 Shape

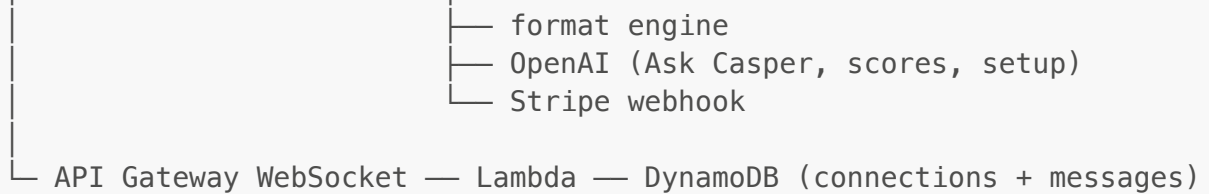
Layer	AWS choice	Why
Client	Next.js PWA on S3 + CloudFront (or Amplify Hosting)	Fast v1. Native wrapper later if iOS push is weak
Auth	Amazon Cognito	Google + Apple IDPs, JWT to APIs, long refresh tokens
API	API Gateway (HTTP + WebSocket) + Lambda (Node.js / TypeScript)	No server to babysit; worldwide later
Rules / formats	Same Lambdas, pure TS format engine	Deterministic; no LLM
AI	Lambda → OpenAI (tools). Key in Secrets Manager	Asked OpenAI, not Bedrock (can add later)
Data	DynamoDB	NoSQL as requested, AWS-native, global tables later
Email / magic link	Amazon SES + Cognito custom challenge (or email OTP)	No password
Files	S3 (avatars, recap images)	
Payments	Stripe → API Gateway webhook → Lambda	49 kr/month, cancel anytime
Secrets / config	Secrets Manager, SSM Parameter Store	
Observability	CloudWatch, X-Ray	
IaC	AWS CDK (TypeScript)	One language; repeatable US/Asia stacks

TBD: TypeScript everywhere — **yes, recommended.**

Not v1 unless we hit a wall: ECS/Fargate (always-on chat), AppSync (alternative to WebSocket API), AWS Bedrock.

10.2 Suggested AWS map





Cognito JWT on every call. Membership (trial / 49 kr / lapsed) stored in DynamoDB (and optionally Cognito attributes). AI Lambdas **refuse** if lapsed — OpenAI is not called.

10.3 DynamoDB (not Mongo)

Fits chat + events + sessions if we **model access patterns**, not one giant document.

Hard limit: 400 KB per item. Do **not** store a whole Play night (all rounds + all chat) in one item.

Sketch (multi-entity, can be single-table later):

Entity	Keys (sketch)	Notes
User	USER#id	membership, trial_start, stripe_customer
Group	GROUP#id	quietMode, sport default
Membership	GROUP#id / USER#id	role: creator, admin, organiser, member
DirectThread	DM#min:max	two user ids
Event	GROUP#id / EVENT#id	bubble, recurrence, RSVP + likes as related items or sets
Session	SESSION#id	organiser, participants, sourceEventId?, sport, format
Match	SESSION#id / MATCH#round#court	sides, score, status
Message	THREAD#... / TS#iso	group, DM, or Play; type text event_bubble table system
Connection	CONN#websocketId	for API Gateway WS fan-out
Invite	token	

Player history: GSI on participantId + sessionId (or write a small PlayerMatch item per result).

Typed Lambda queries only — the model never invents DynamoDB queries. Tools like `get_player_history` we implement.

Later worldwide: DynamoDB **global tables**, API Gateway + Lambda in `us-east-1` / `ap-southeast-1`, same table. Cognito: user pool per region or a planned multi-region auth story (decide when we leave Nordics).

10.4 AI architecture

```

Ask Casper / Create Event / Play setup / "Court 1 11-8"
  ↓
API Gateway → Lambda
  ↓
Authz: Cognito JWT + role + trial/paid
  ↓
Load current session into the prompt
(matches on screen: ids, court, names, status)
  ↓
OpenAI tools → create_event | start_session | record_result | swap_player |
...
  ↓
Format engine in Lambda (validate match id, write DynamoDB)
  ↓
WebSocket fan-out: table + next games

```

Play score path: organiser talks to Casper while the round is visible. System prompt includes **current open matches** with stable ids. Tool `record_result({ matchId, scoreA, scoreB })` is rejected if `matchId` is not open. Casper then confirms in chat ("Court 1, Cassie/Erik 11-8 Anna/Moa").

Connect: no LLM on member messages, RSVP, or likes.

Rules:

- Allowed tools + format codes only
- `sport` passed in (`pickleball`)
- Idempotent scores
- Organiser undo last score
- Lapsed membership → no OpenAI call

Prefer **gpt-4o-mini** (or similar) for score registration; a stronger model only for first-time Play setup if needed.

10.5 What the AI is allowed to do vs not

Allowed (organiser + membership)	Not allowed
Ask clarifying questions after Ask Casper / Create Event	Run on every Connect group message
Recommend a format from the catalogue	Invent a rotation system
<code>record_result</code> on a match id we provided	Invent a match / wrong court
<code>create_event</code> , <code>start_session</code> , <code>swap_player</code>	Scan DynamoDB ad hoc
Summarise who's in / a table	Use training data as history
Draft share text	Post into WhatsApp

10.6 Data the AI needs from us

- Current open matches (ids, court, names) — **required for scores**

- Group members, quietMode, event roster / on-site
- Format catalogue + **sport**
- Standings snapshot
- Aliases
- Caller membership status

10.7 Auth (Cognito)

- **Sign in with Google** (Cognito federated)
- **Sign in with Apple** (Cognito federated; needed for later iOS)
- **No passwords**
- **Long-lived login** (Cognito refresh tokens, 30 days+ as we configure)
- **Magic link** if session gone or new device: SES email + one-time token / Cognito custom auth (email OTP is an acceptable v1 stand-in)

Walk-up: organiser adds a display name in Play; claim later. **TBD:** merge.

10.8 Realtime and push

- **Chat / RSVP / table:** API Gateway **WebSocket**, connection ids in DynamoDB, Lambda fan-out to the thread
- **Push:** web push via a Lambda later; **iOS PWA push is weak** — Pinpoint/native wrapper when Connect must ping like WhatsApp
- Quiet groups: still push **event reminders** and admin posts; do not notify on member chat that is disabled anyway

10.9 Why not the old Docker/Fastify/Mongo suggestion

That matches other personal apps, not this one. Casper needs **Cognito, WebSockets, NoSQL, worldwide**. DynamoDB + Lambda is the AWS version of that. If WebSocket + Lambda gets painful in practice, **API Gateway WS stays** and we can move the format engine to **Fargate** without leaving AWS.

11. Information architecture

```

User (Cognito sub + DynamoDB profile: trial | paid 49kr | lapsed | never)
├── DirectThread (1:1)                                [Connect] always on
├── Group (quietMode on/off, roles)                     [Connect]
│   ├── Admin/member messages (members blocked if quiet)
│   └── Event bubbles (RSVP + like always)
│       └── Start Play → Session
User starts Play from scratch
├── Session (sourceEventId: null)                       [Play]
│   ├── organiserId (always)
│   ├── participantIds (can view)
│   ├── Match items (not one blob)
│   └── Messages (score confirmations, tables)

```

12. Phased build

Phase 0 — this document

Agree product + AWS map. No code.

Phase 1 — thin vertical slice (CDK, one region)

- Cognito: Google + magic link/OTP (Apple can follow)
- One group, quiet toggle, Create Event bubble, RSVP + like
- Ask Casper only on organiser actions
- Play from bubble **and scratch**
- One format (Americano or round robin)
- **AI register score** to match on screen + engine table
- DynamoDB + WebSocket updates
- Share: copy link
- Membership stub (everyone on fake trial)

Out of scope: Stripe, DUPR, native, voice, DM polish.

Phase 2 — real group night

- Invites, roles, weekly recurrence
- Private chats
- More formats
- SES invites, web push
- Walk-up names
- Recap on S3

Phase 3 — productize

- Stripe **49 kr/month**, 3-month trial, cancel anytime
- Apple Sign-In + iOS push story
- DUPR export
- Multi-region when US/Asia is real (**sport** expansion if demanded)

13. Risks

Risk	Mitigation
49 kr eaten by fat prompts on every score	Mini model + only current matches in context
Player scores still ambiguous	Keep question open; AI path is organiser-only for now
Quiet groups feel dead	RSVP, likes, admin posts, DMs still alive
DynamoDB 400 KB / awkward queries	Split matches & messages; GSIs for history

Risk	Mitigation
Lambda + WebSocket complexity	Standard connection-table pattern; Fargate fallback
Cognito magic link is fiddly	Email OTP as v1
iOS PWA push	Native/Pinpoint later
Trial abuse	One trial per Cognito user
Name collision	Check domains/trademarks
DUPR TOS	Manual export first

14. Open questions

Product

1. **Player scores:** AI-only (organiser), or players may tap/type too? **Kept open.**
2. Quiet mode: is member chat a **hard block**, or a warning / slow mode? (Lean hard block.)
3. Ask Casper from **private chats**, or only groups + Play?
4. After Play ends: keep forever vs archive after N days?
5. Walk-up names: guests forever vs require claim?
6. Language: Swedish UI vs English-first?
7. Voice in v1 or text only?
8. Two Play sessions at once from one group?

Formats

9. First format: Americano vs round robin?
10. Singles in v1, or doubles only?
11. Bye rules for odd numbers?

Business

12. Two admins = 2 × 49 kr, or a **group seat** later?
13. Trial once; cancel day 89 and return?
14. VAT / Stripe vs app stores when we leave web-only?

Tech

15. HTTP API + WebSocket vs **AppSync** subscriptions?
16. Next on S3/CloudFront vs **Amplify Hosting** vs SST/OpenNext SSR?
17. Domain name besides Casper?
18. Push: PWA vs native shell sooner?
19. Email OTP vs true magic link in Cognito v1?

15. Decisions already made

From the original brief:

- Working name: Casper
- Chat-first, OpenAI
- Connect + Play
- Last-minute on-site roster is required
- AI interviews the organiser for courts, names, DUPR yes/no
- Results + tables in the chat
- History in NoSQL
- Formats locked in backend API; AI may call them, not invent them
- Web/PWA first

From brainstorm 2:

- **Sport:** pickleball for now; formats reusable later
- **Two tabs**, both chat
- **Groups:** create / join / invite
- **Auth:** Google or Apple, no password, long-lived login, magic link fallback
- **Own chat**, share out, no WhatsApp bridge

From brainstorm 3:

- **Who organises:** right **role and membership**
- **Who pays:** the one who **uses AI**. Players **read** free
- **Plan:** 3 months free, then monthly, cancel anytime
- **Connect Casper:** Ask Casper or Create Event
- **Groups + private chats; events are bubbles;** weekly groups; quiet option
- **Scratch Play allowed;** organiser + participants can view

From brainstorm 4:

- **Quiet mode:** creator on/off per group. **Members still RSVP, like events, and use private chats. Creator/admin can always post.** Member free-chat in that group is what quiet turns off.
- **Scores:** organiser **asks AI** to register the score on the **match shown on screen**. Player tap vs AI-only **kept open**.
- **Price:** 49 kr/month (working)
- **Stack:** AWS worldwide path — Cognito, Lambda, API Gateway, DynamoDB, S3/CloudFront, SES, CDK. Not Mongo/Docker-as-default.

16. Next step

Useful next: first format (Americano vs RR), Swedish vs English UI, and whether Phase 1 is CDK in eu-north-1 with a fake trial. Then a later `implementation-plan.md` for the first repo.